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Electric Vehicle
Codec Technologies India
Internship 2026

EV Market Forecast Dashboard

Dash / Plotly dashboard on real IEA Global EV Data Explorer
2026 data with logistic, Bass and trend forecasts to 2035

AT A GLANCE

52,800

IEA rows downloaded (real)

0.53 M -> 21 M

World EV sales 2015 -> 2025

25 % / 53 % / 4 %

World share 2025 / China / India

66.5 / 91.4 %

World logistic L=100 % 2030 / 2035

27.7 %

World free-fit ceiling (L)

53 %

IEA STEPS 2035 benchmark, World

20 (6 weakening, all 2023-25)

Policy events in register

38 / 40

Tests re-run here

VERDICT

TESTS 40/40

38 RE-RUN + 2 RECORDED

REAL IEA DATA

CC BY 4.0

DASH NOT RUN HERE

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1. Objective and brief mapping

Brief: an interactive dashboard to predict EV market growth using Python (Pandas, Plotly, Dash), analysing global trends, government policies and adoption rates with graphs, maps and growth models. I built a Dash app with Overview / Forecast / Map / Policies / Data & Sources tabs, a models module that fits logistic (free and fixed ceiling), Bass diffusion, CAGR, exponential and linear models to the EV share of new car sales for World, China, Europe, USA and India and forecasts to 2035, a choropleth of the 2025 share in 61 countries, and a timeline of 20 real policy events.

All historical numbers are real IEA data: 52,800 rows downloaded with curl from the public API behind the IEA Global EV Data Explorer (GEO 2026 edition, api.iea.org/evs?year=YYYY, one call per year 2010-2035, on 2026-09-02, CC BY 4.0) and concatenated unchanged. The policy register (data/policies.csv) was hand-compiled with an official or trade-press source URL per row. A test cross-checks the API series against the GEO 2026 executive summary (World 21.0 M / 25 %, China 53 %, Europe 28 %, USA 10 %).

2. Model and architecture

src/load_data.py reads the tidy CSVs; src/models.py fits with `scipy.optimize.curve_fit` on 2015-2025 share $s(t)$ in percent - logistic $s = L / (1 + e^{-(k(t-t_0))})$ with L free in $[\max s, 100]$ or fixed by the ceiling slider; Bass $s = m F(\tau)$, $F = (1 - e^{-(p+q)\tau}) / (1 + (q/p) e^{-(p+q)\tau})$, $\tau = t - 2014$; CAGR, exponential and linear clipped to 0-100 %. Confidence band = parametric bootstrap from the `curve_fit` covariance (400 draws, SE damped $\times 0.5$) for the S-curves, $\pm 1.645 \text{ RMSE} \sqrt{1 + \text{horizon}}$ otherwise. EV units = share $\times$ total market grown at the 2015-2025 implied rate capped to $[-1, +3] \%$ /yr. Validation: in-sample RMSE / R^2 plus a hold-out fit on 2015-2022 scored on 2023-2025.

S-curve fits by region — Logistic S-curve (fixed ceiling)

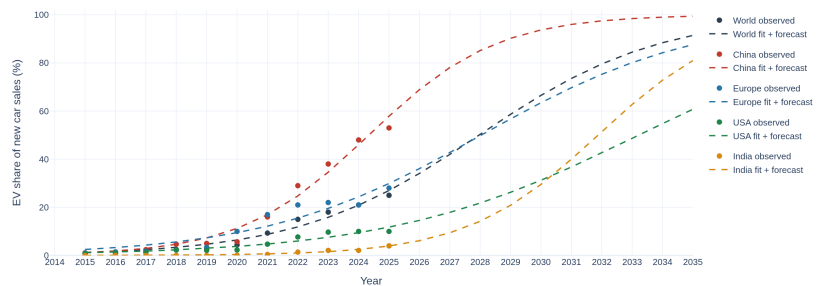


Fig. 1 - Logistic fits with the ceiling fixed at 100 % for the five regions (kaleido PNG from the project's recorded run).

3. Method and implementation

- Fitted parameters: World logistic free L 27.7 %, k 0.676, t_0 2022.0; fixed $L = 100$: k 0.336, t_0 2028.0; Bass m 27.8 %, p 0.0031, q 0.667. China free L 58.5 %, k 0.810, t_0 2022.2; fixed k 0.477, t_0 2024.3. Europe L 28.0 %, fixed t_0 2028.0. USA L 11.6 %, fixed t_0 2033.2. India L 9.2 %, k 0.580, t_0 2025.6; Bass m 10.2 %, q 0.556 (lowest); 5-yr share CAGR +101.6 %/yr.
- Policy-scenario slider (-1 ... +1) scales the growth rate by $(1 + 0.3 f)$ and the saturation by $(1 + 0.1 f)$, re-anchored at the 2025 observation - a what-if lever, not a calibrated elasticity. Saturation-ceiling slider sets L for the fixed logistic.
- src/export_static.py writes results/dashboard_static.html (self-contained, bundled topojson) and 15 of 16 kaleido PNGs (the choropleth PNG failed because kaleido's Chromium could not fetch plotly's topojson in the sandbox; it renders in the app and static HTML).

4. Results (results/forecast_table.csv, model_params.json)

Goodness of fit, RMSE in percentage points (in-sample / hold-out 2023-25): World logistic free 0.73 / 16.8, fixed 1.64 / 16.8, Bass 0.75 / 16.3, CAGR 2.85 / 24.1, linear 2.74 / 7.0; China logistic 1.51 / 17.6; Europe logistic 1.83 / 2.9 and linear 2.88 / 2.0 (best hold-out); USA logistic 0.72 / 7.3, linear 1.22 / 2.6; India logistic 0.26 / 20.2, Bass 0.26 / 2.5. Fits on 2015-2022 over-predicted 2023-25 by 7 pp for the USA and 17 pp for the world. Market facts from the data: world electric car sales 0.53 M (0.68 %) in 2015 -> 21 M (25 %) in 2025, +24 % over 2024's 17 M; stock 75 M (5 % of cars on the road); China 13 M of the 21 M (62 %), Europe 4.2 M, USA 1.5 M (flat since 2024), India 0.17 M (4 %, from 2.1 % in 2024).

Region	2025 actual	Logistic L=100: 2030 / 2035	Logistic free 2030 / 2035	Bass 2030 / 2035	Linear 2030 / 2035	IEA STEPS 2035
World	25 %	66.5 / 91.4 %	27.6 / 27.7 %	27.6 / 27.8 %	34.9 / 47.8 %	53 %
China	53 %	93.7 / 99.4 %	58.4 / 58.5 %	58.3 / 58.4 %	74.2 / 100 %	94 %
Europe	28 %	63.5 / 87.6 %	28.0 / 28.0 %	28.0 / 28.0 %	41.3 / 56.0 %	85 %
USA	10 %	31.3 / 60.8 %	11.6 / 11.6 %	11.4 / 11.5 %	15.7 / 21.2 %	20 %
India	4 %	29.5 / 81.0 %	8.5 / 9.1 %	9.3 / 10.2 %	4.3 / 6.1 %	24 %

Region	Logistic free (L %, k, t0)	Logistic L=100 (k, t0)	Bass (m %, p, q)	5-yr share CAGR	In-sample RMSE logistic free / Bass (pp)
World	27.7, 0.676, 2022.0	0.336, 2028.0	27.8, 0.0031, 0.667	+41.6 %/yr	0.73 / 0.75

KEY TAKEAWAY Eleven years of real IEA data fit an S-curve well but cannot fix its ceiling - a free fit says 28 % world share for ever, a 100 % ceiling says 91 % by 2035, IEA STEPS sits at 53 % - so the dashboard exposes the ceiling as a slider.

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Region	Logistic free (L %, k, t0)	Logistic L=100 (k, t0)	Bass (m %, p, q)	5-yr share CAGR	In-sample RMSE logistic free / Bass (pp)
China	58.5, 0.810, 2022.2	0.477, 2024.3	58.4, 0.0011, 0.811	+56.2 %/yr	1.51 / 1.53
Europe	28.0, 0.720, 2020.9	0.281, 2028.0	28.0, 0.0056, 0.703	+22.9 %/yr	1.83 / 1.83
USA	11.6, 0.649, 2021.3	0.245, 2033.2	11.5, 0.0053, 0.666	+34.2 %/yr	0.72 / 0.74
India	9.2, 0.580, 2025.6	0.464, 2031.9	10.2, 0.0007, 0.556	+101.6 %/yr	0.26 / 0.26

World: EV sales share, Logistic S-curve (fixed ceiling) — L=100, k=0.336, t0=2028

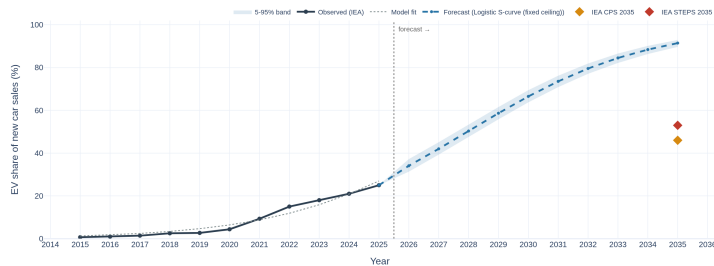


Fig. 2 - World EV share forecast to 2035 with the bootstrap 5-95 % band and the IEA STEPS 2035 benchmark.

Global electric car sales by region (IEA, historical)

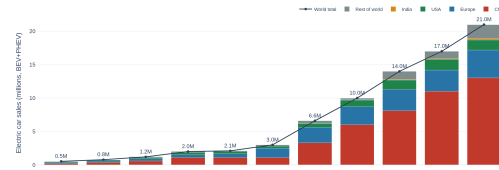


Fig. 3 - Real IEA data: electric car sales by region 2010-2025, 0.53 M in 2015 to 21 M in 2025 with China at 13 M.

5. Verification

The README records 40 pytest checks passing. Here pytest is not installed and **plotly / dash are unavailable**, so I re-ran the suite with a fixture/parametrize-aware runner in which figures, plotly and dash are stubbed: **38 passed, 0 failed** in 0.7 s; the **2 figure / app tests were not re-run** and their result is the project's recorded run (2 passed). The Dash server and Playwright screenshots were not repeated.

Test	Checks	Result here
test_region_table_loads_without_nan_in_key_columns, test_country_map_table, test_headline_numbers_match_iea_geo2026_summary	five modelling regions complete 2015-2025; 61-country map table with ISO3; API series matches the GEO 2026 executive summary	3 PASS
test_policies_have_sources_and_years, test_sources_csv_covers_every_data_file	every policy row has a year and a source URL; sources.csv registers every data file	2 PASS
test_scurve_fit_share_within_0_100[5 regions x 3 models], test_scurve_forecast_is_monotonic[5 x 3]	fitted shares stay within 0-100 %; S-curve forecasts are monotonic for every region and model	30 PASS
test_policy_scenario_shifts_forecast_in_right_direction, test_cagr_and_simple_models_run_and_are_capped, test_fit_all_writes_results	+f raises and -f lowers the forecast; CAGR / exponential / linear capped at 100 %; fit_all writes forecast_table.csv and model_params.json	3 PASS
test_all_static_figures_build, test_app_layout_builds	all Plotly figures build with data; Dash layout, tab renderers and forecast callback run without a server	2 NOT RE-RUN (plotly/dash); recorded PASS

6. Data integrity and declared substitutions

- Real data, declared limits: IEA values are rounded to 2 significant figures; implied_total_car_sales = ev_sales / share is my derived quantity; IEA 'Europe' = EU27 + EFTA + UK + Türkiye / Balkans; a few small countries lack early years but the five modelling regions have no gaps.
- Eleven annual observations cannot identify the saturation level - the free-fit ceilings mean 'what the last two years imply', not a prediction, which is why the ceiling is a slider. The policy slider mapping (+/-30 % growth, +/-10 % saturation) is my assumption. Unit forecasts assume the total market grows at the historical implied rate.
- Not run here: the Dash server, Playwright screenshots (results/screenshots/ are from the recorded run) and kaleido PNG export; no statsmodels dependency was needed.

7. Learnings

The global market grew 40x in a decade and China is 62 % of it, already past the S-curve midpoint (t0 2022-2024, k 0.5-0.8/yr). Europe's 2024 stall (22 -> 21 %) coincides with Germany ending the Umweltbonus and the 2025 jump to 28 % with the EU 2025 CO2 targets; the USA is the outlier at a flat 10 % with the 30D credit ending and the ACC II waiver revoked. Of 20 policy events, all 6 'weakening' ones are in 2023-2025 and in the West, while China, India, Thailand and the UK kept adding support - which is why I would not trust a single global S-curve and model regions with their own ceilings instead.

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